SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DATA HANDLING AND VISUALIZATION -DSA0604

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QUESTION 1: Chart families and the base R plotting model

AIM: To construct a layered plot using base R, identify suitable chart families for different analytical questions, and compare a pie chart with a sorted bar chart.

a) Layered base R plot

PROCEDURE:

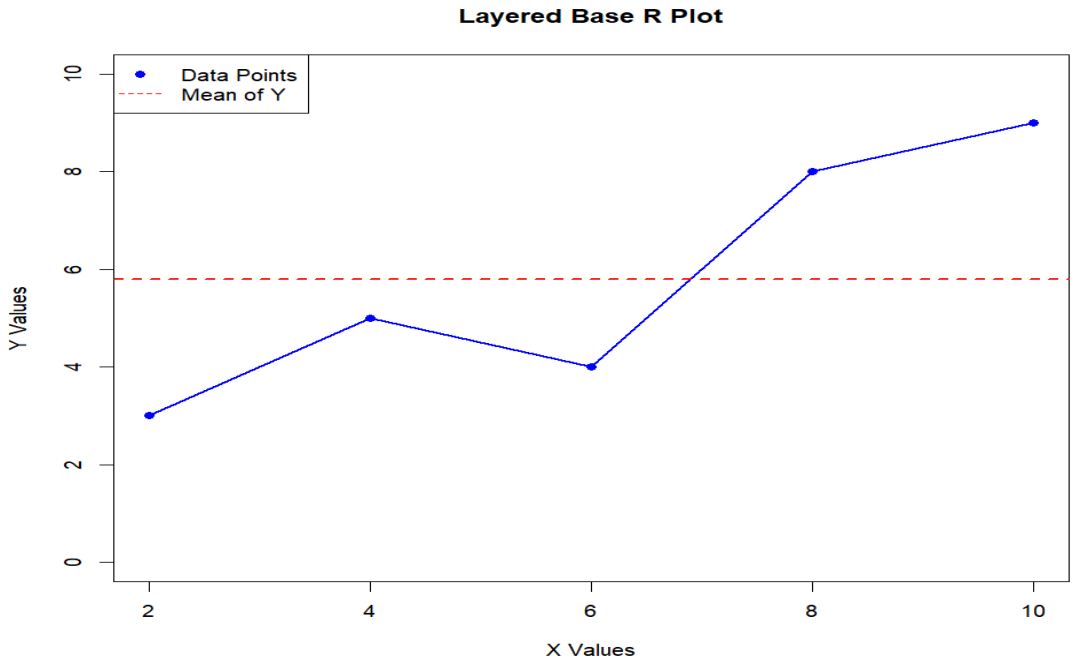
1. Define the vectors x and y using the values given in the question.
2. Open an empty plotting canvas with plot.new() and plot.window().
3. Add the axes and axis labels using axis() and title().
4. Add data points with points() and connect them using lines().
5. Calculate mean(y) and draw a dashed horizontal reference line using abline().
6. Add a legend and save the plot as layers.png using png() and dev.off().

CODE:

```
  # Define the data
x <- c(2, 4, 6, 8, 10)
y <- c(3, 5, 4, 8, 9)
# Calculate mean
m <- mean(y)
# Function to draw the plot
draw_plot <- function() {
 # Open empty canvas
 plot.new()
 # Define plotting window
 plot.window(xlim = range(x), ylim = c(0, 10))
 # Add axes
 axis(1)
 axis(2)
 box()
 # Add title and labels
 title(main = "Layered Base R Plot",
       xlab = "X Values",
       ylab = "Y Values")
Add points
 points(x, y, col = "blue", pch = 19)
 # Add connecting line
 lines(x, y, col = "blue", lwd = 2)
 # Add mean line
 abline(h = m, col = "red", lty = 2, lwd = 2)
 # Add legend
 legend("topleft",
        legend = c("Data Points", "Mean of Y"),
        col = c("blue", "red"),
        pch = c(19, NA),
        lty = c(NA, 2))
```

```
}
# Display graph in RStudio
draw_plot()
# Save graph as PNG
png("layers.png", width = 800, height = 600)
draw_plot()
dev.off()
# Display mean
    cat("Mean of Y =", m)
```

OUTPUT / OBSERVATION



b) Select the chart family

Question	Chart family	Suitable chart
What share of the monthly budget goes to rent?	Part-to-whole	Pie chart or 100% stacked bar
How are the marks of 60 students spread?	Distribution	Histogram
How did website visits change over 12 months?	Time series	Line chart
Is height related to weight?	Relationship	Scatter plot
Which students are friends with each other?	Relationship / network	Network graph

c) Pie chart or sorted bar chart for budget shares

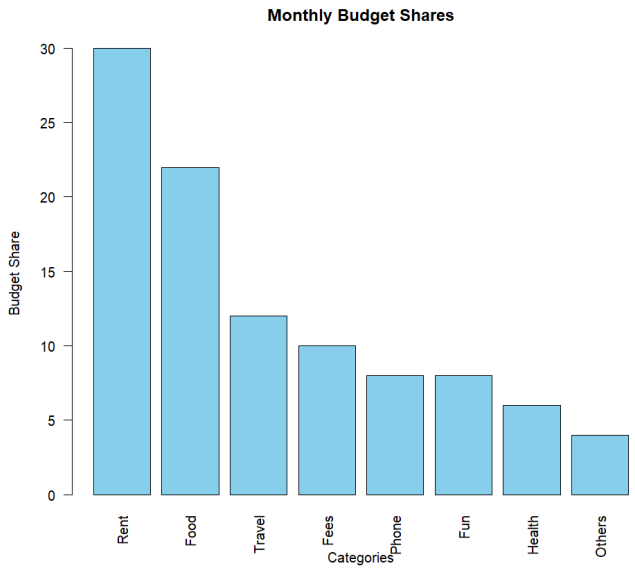
The budget values are Rent 30, Food 22, Travel 12, Fees 10, Phone 8, Fun 8, Health 6 and Others 4.

```
budget <- c(Rent = 30, Food = 22, Travel = 12,
            Fees = 10, Phone = 8, Fun = 8,
            Health = 6, Others = 4)
```

```
# Recommended: sorted bar chart
budget_sorted <- sort(budget, decreasing = TRUE)
barplot(budget_sorted, las = 2, col = "steelblue",
        main = "Monthly Budget Shares",
        ylab = "Share")
```

A sorted bar chart is preferable because it makes comparisons and ranking easier, especially when there are several categories and some values are close. A pie chart communicates part-to-whole composition but small slices such as Health and Others are harder to compare accurately.

OUTPUT / OBSERVATION:



RESULT: The layered base R plot was constructed, appropriate chart families were identified, and a sorted bar chart was selected for comparing the budget categories.

QUESTION 2: Connecting and exploring data in Tableau

AIM: To connect the Orders sheet in Tableau, prepare field roles, calculate Profit Ratio, inspect row-level data and evaluate the Profit field.

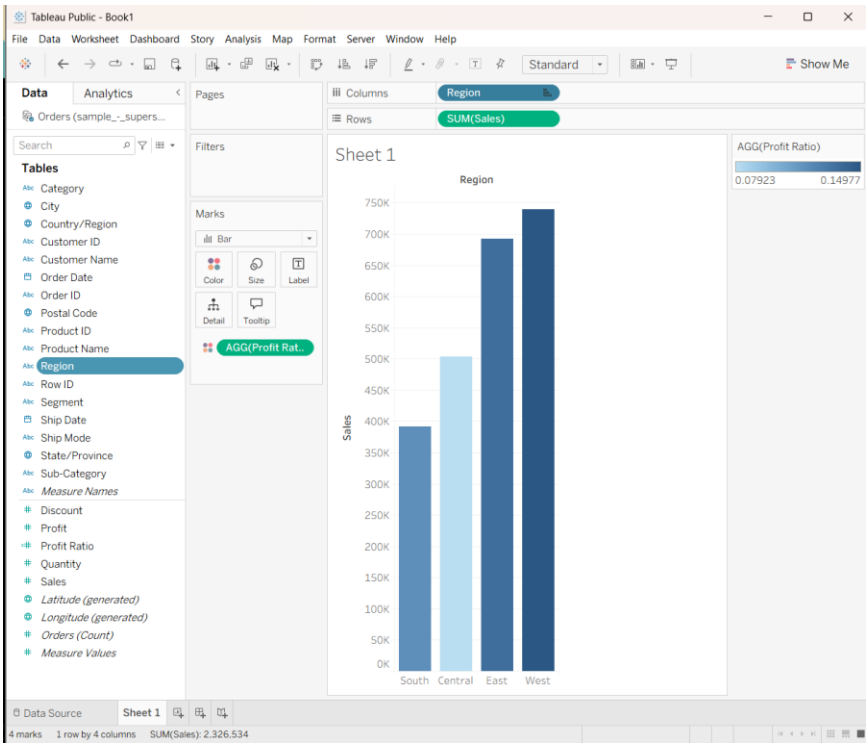
a) Connect data and create Sales by Region coloured by Profit Ratio

PROCEDURE:

- Open Tableau and choose Microsoft Excel under Connect.
- Select Retail_India_Tableau_Practice.xlsx and open the Orders sheet.
- In the Data Source page, check that fields have the correct data types.
- Right-click Row ID, Customer No and Order ID and choose Convert to Dimension.
- Create a calculated field named Profit Ratio with the formula below.
- Open a worksheet. Drag Region to Columns and Sales to Rows.
- Set the Marks type to Bar. Sort Region in descending order by SUM(Sales).
- Drag Profit Ratio to Color. Format it as a percentage if required.

```
SUM([Profit]) / SUM([Sales])
```

OUTPUT / OBSERVATION: A sorted bar chart displays total Sales for each Region. Colour represents Profit Ratio. Record the region-wise Sales and Profit Ratio shown in your Tableau view; these values depend on the supplied workbook.



b) Explain why 2,534 rows exceed the number of distinct orders

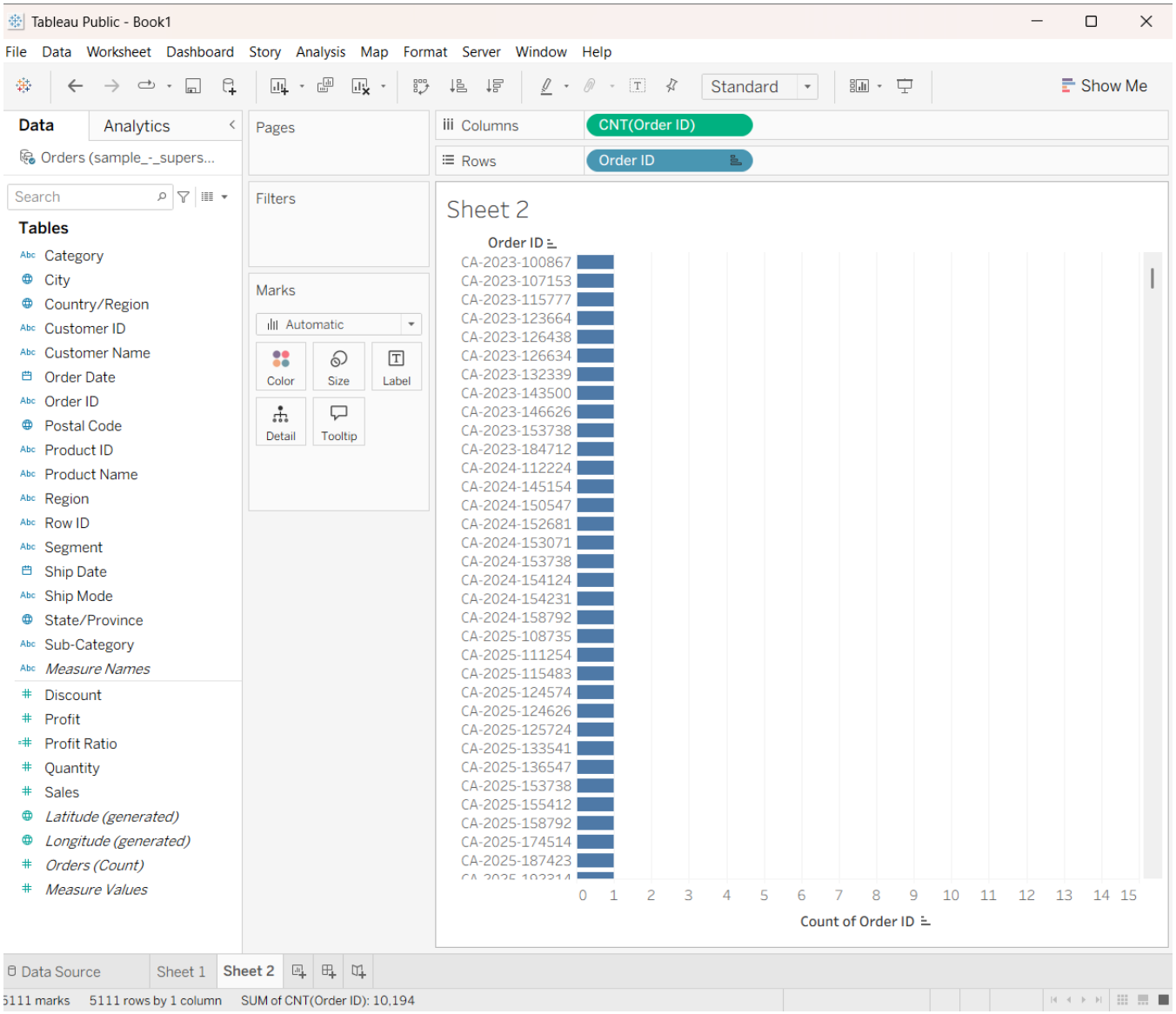
The Orders sheet is at line-item level: one order can contain multiple product/order lines. Therefore, the number of rows can be greater than the number of distinct Order ID values. Use COUNTD([Order ID]) to calculate distinct orders.

7. Open a new worksheet and drag Order ID to Text.
8. Change its aggregation to Count (Distinct), or create COUNTD([Order ID]).
9. Compare the distinct order count with the total row count (Number of Records).

10. To find an example, place Order ID and product/item-level fields in a text table, then filter to an Order ID that appears on more than one row.

```
COUNTD([Order ID])
```

OUTPUT / OBSERVATION:



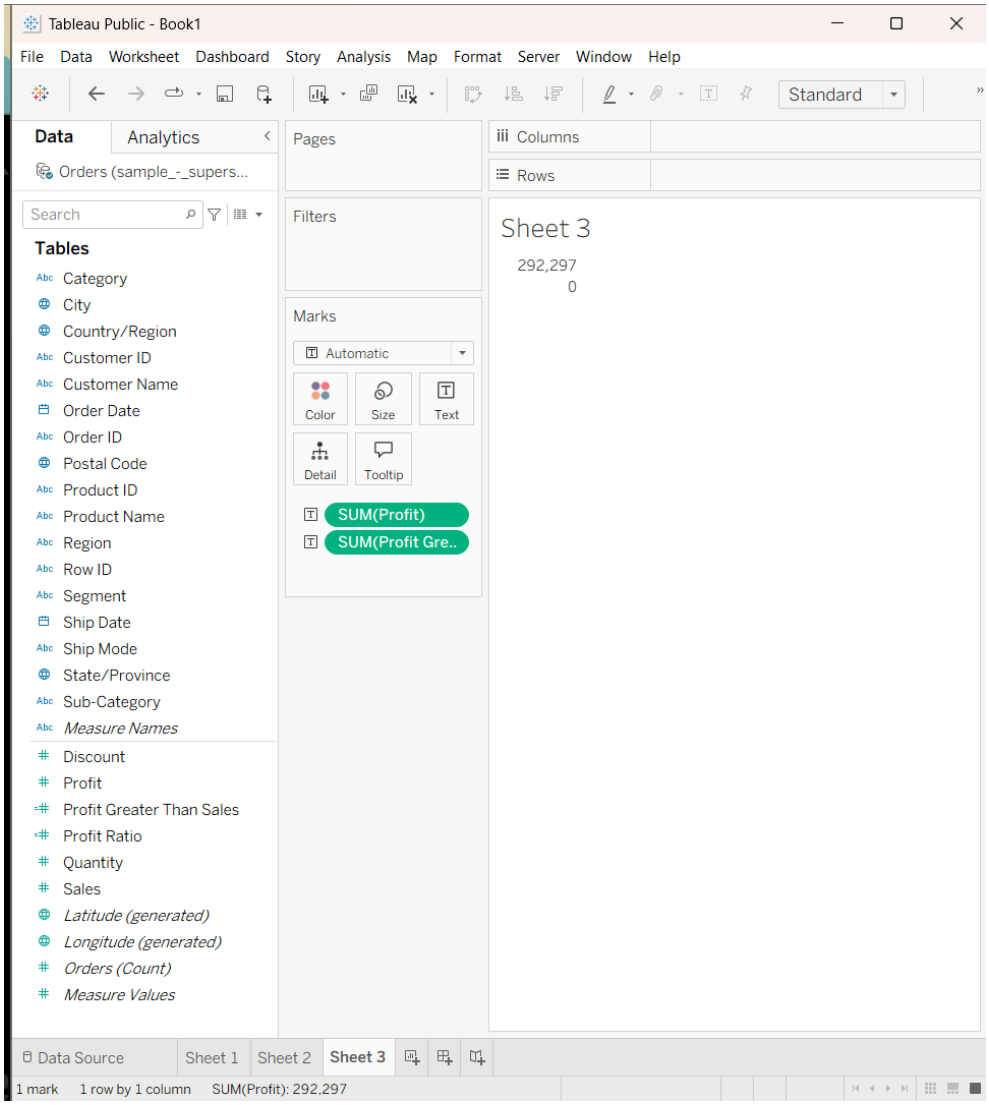
c) Check Profit field quality

Create a Boolean check for lines where Profit exceeds Sales and inspect the minimum Profit value.

```
[Profit] > [Sales]
```

11. Create the calculated field Profit Greater Than Sales using the formula above.
12. Place it on Filters or Text and count the True records to obtain the number of lines where Profit is greater than Sales.
13. Drag Profit to Text and change aggregation to Minimum to identify the largest loss (the most negative Profit value).
14. Inspect unusual records and confirm the underlying Sales and Profit values before reporting totals.

OUTPUT / OBSERVATION:



RESULT: The Orders data connection, dimensions, Profit Ratio calculation and quality-check workflow were specified. Actual data-dependent values must be recorded from Tableau.

QUESTION 3: Customizing a base R plot

AIM: To create a customized scatter plot in base R and understand plotting symbols, symbol size and axis limits.

a) Scatter plot of weight against height

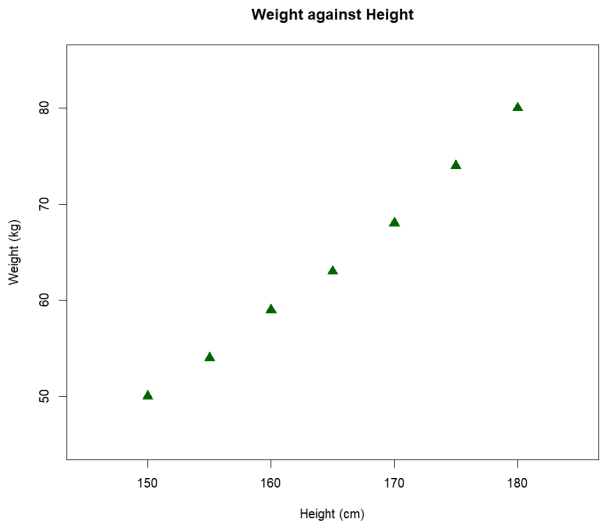
PROCEDURE:

- Define the height and weight vectors.
- Use plot(weight ~ height) to plot weight against height.
- Set pch = 17, col = darkgreen and cex = 1.5.
- Set xlim to 145–185 and ylim to 45–85.
- Add suitable axis labels and a title.

```
height <- c(150, 155, 160, 165, 170, 175, 180)
weight <- c(50, 54, 59, 63, 68, 74, 80)

plot(height, weight,
     pch = 17, col = "darkgreen", cex = 1.5,
     xlim = c(145, 185), ylim = c(45, 85),
     xlab = "Height (cm)", ylab = "Weight (kg)",
     main = "Weight against Height")
```

OUTPUT / OBSERVATION:



b) Meaning of pch and cex

Setting	Meaning	Appearance / effect
pch = 1	Open plotting symbol	Open circle
pch = 17	Triangle plotting symbol	Filled triangle
pch = 19	Solid plotting symbol	Filled circle
cex = 0.5	Character expansion factor	Symbols are half the default size
cex = 2	Character expansion factor	Symbols are twice the default size

c) Evaluate ylim = c(0, 100)

The given weight values range from 50 kg to 80 kg.

1. Using ylim = c(0, 100):

- Creates unnecessary empty space in the graph.
- Makes the plotted points appear compressed.
- Reduces the visibility of differences between weight values.

2. Using ylim = c(45, 85):

- Focuses on the actual data range.
- Improves the visibility of plotted points.
- Makes comparison easier and clearer.

R Code:

```
height <- c(150, 155, 160, 165, 170, 175, 180)

weight <- c(50, 54, 59, 63, 68, 74, 80)

# Plot with ylim 0 to 100

plot(height, weight,

    pch = 17, col = "darkgreen",

    ylim = c(0, 100),

    main = "Y-axis Limits (0 to 100)",

    xlab = "Height (cm)",

    ylab = "Weight (kg)")

# Plot with ylim 45 to 85

plot(height, weight,

    pch = 17, col = "darkgreen",

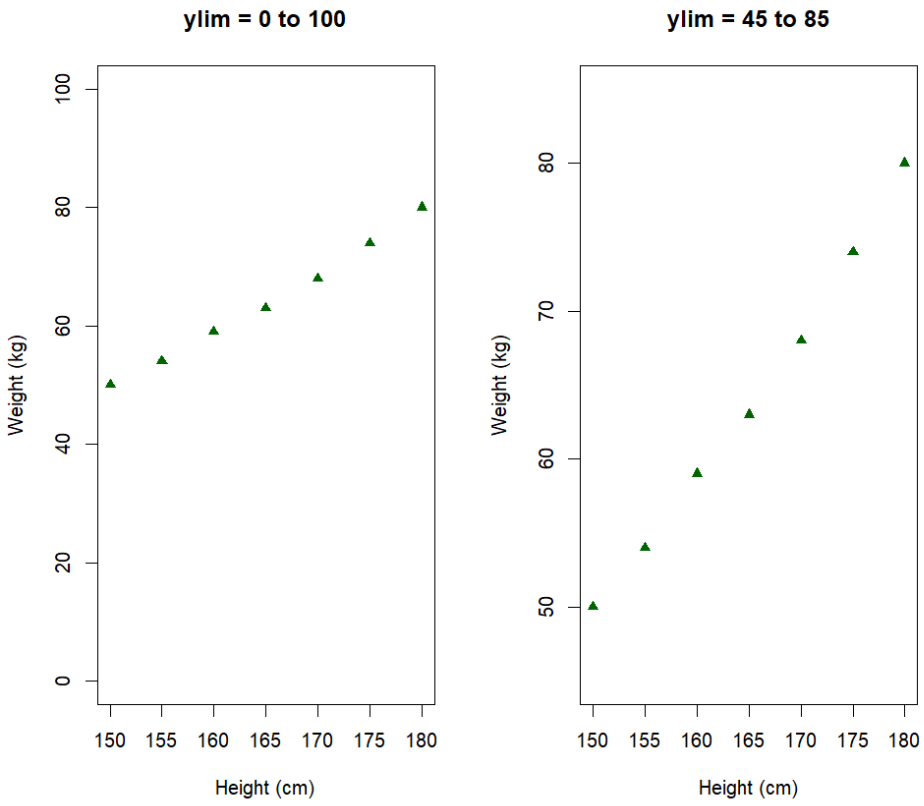
    ylim = c(45, 85),

    main = "Y-axis Limits (45 to 85)",

    xlab = "Height (cm)",

    ylab = "Weight (kg)")
```

OUTPUT / OBSERVATION:



RESULT: The customized scatter plot was created and the effects of pch, cex and ylim were explained.

QUESTION 4: Layers, text and saving a figure

AIM: To create a histogram, add mean and median reference lines, use a legend and save the figure as a PDF.

a) Histogram with mean and median

PROCEDURE:

- Define the scores vector.
- Open the PDF graphics device using pdf().
- Draw the histogram using hist() with breaks = 5.
- Calculate mean and median of the scores.
- Add vertical reference lines with abline(v = ...).
- Add a legend and close the graphics device with dev.off().

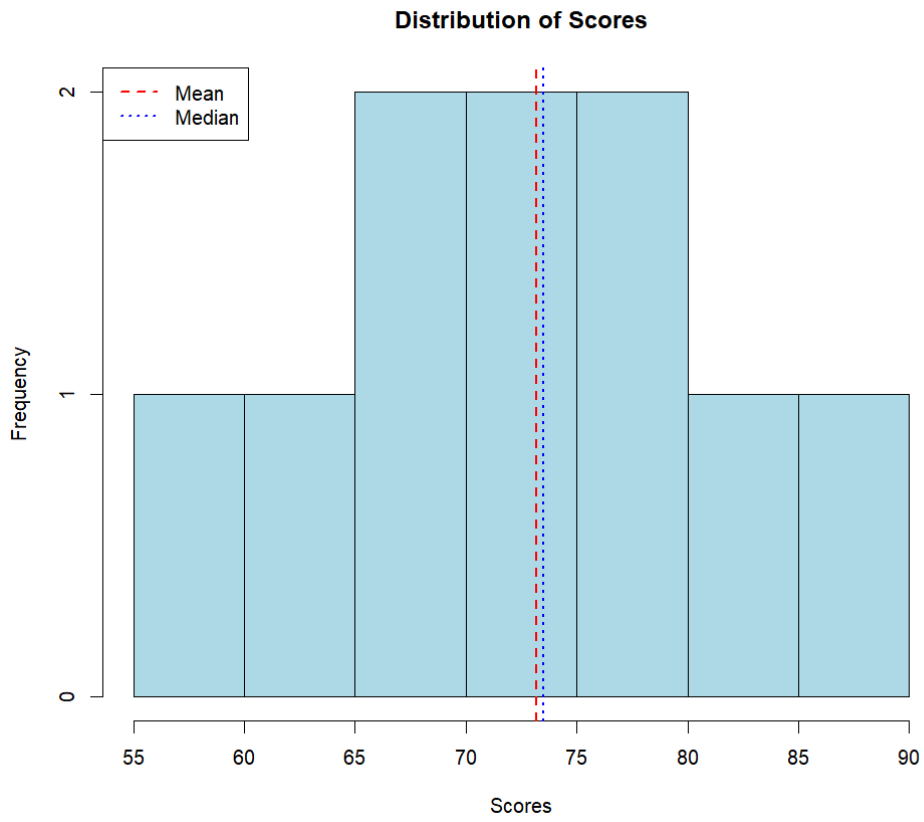
```
scores <- c(55, 62, 68, 70, 72, 75, 78, 80, 84, 88)

mean_score <- mean(scores)
median_score <- median(scores)

pdf("scores.pdf", width = 7, height = 5)
hist(scores, breaks = 5, col = "lightblue",
    border = "black",
    main = "Distribution of Scores",
    xlab = "Scores")
abline(v = mean_score, col = "red", lwd = 2, lty = 2)
abline(v = median_score, col = "blue", lwd = 2, lty = 3)
legend("topleft",
      legend = c(paste("Mean =", round(mean_score, 2)),
                 paste("Median =", median_score)),
      col = c("red", "blue"), lty = c(2, 3),
      lwd = 2, bty = "n")
dev.off()

mean_score
median_score
```

OUTPUT / OBSERVATION:



b) Order of high-level and low-level plotting functions

hist() is a high-level plotting function: it creates a new plot and establishes the plotting region. abline() is a low-level function: it adds a line to an existing plot. If abline() is called before hist(), there may be no active plotting region, so the line cannot be added correctly (or a plot-related error can occur). The correct order is to call hist() first and then abline().

c) PNG or PDF

PDF is generally better for a printed report because it is a vector format and can be enlarged without losing sharpness. PNG is generally better for a web page because it is a widely supported raster image format. Use high resolution when saving PNG images.

OUTPUT / OBSERVATION: The histogram is saved as scores.pdf with mean and median lines and a legend.

RESULT: The histogram was generated, reference lines and a legend were added, and the figure was saved in PDF format.

QUESTION 5: Choosing the chart for the question

AIM: To construct a pie chart, a monthly temperature line chart and a height-weight scatter plot in one graphics layout, and select appropriate chart families.

a) Three charts side by side

PROCEDURE:

- Define budget, temp, height and weight using the given values.
- Set the graphics layout to one row and three columns using par(mfrow = c(1, 3)).
- Draw the budget pie chart.
- Draw monthly temperature using plot(type = 'o') to show points connected by a line.
- Draw the scatter plot of weight against height.
- Reset the graphics layout after plotting if continuing with other plots.

```
budget <- c(Rent = 40, Food = 25, Travel = 15, Savings = 20)
temp <- c(28, 30, 33, 35, 34, 31, 29, 27, 26, 25, 24, 26)
height <- c(150, 155, 160, 165, 170, 175, 180)
weight <- c(50, 54, 59, 63, 68, 74, 80)

par(mfrow = c(1, 3))

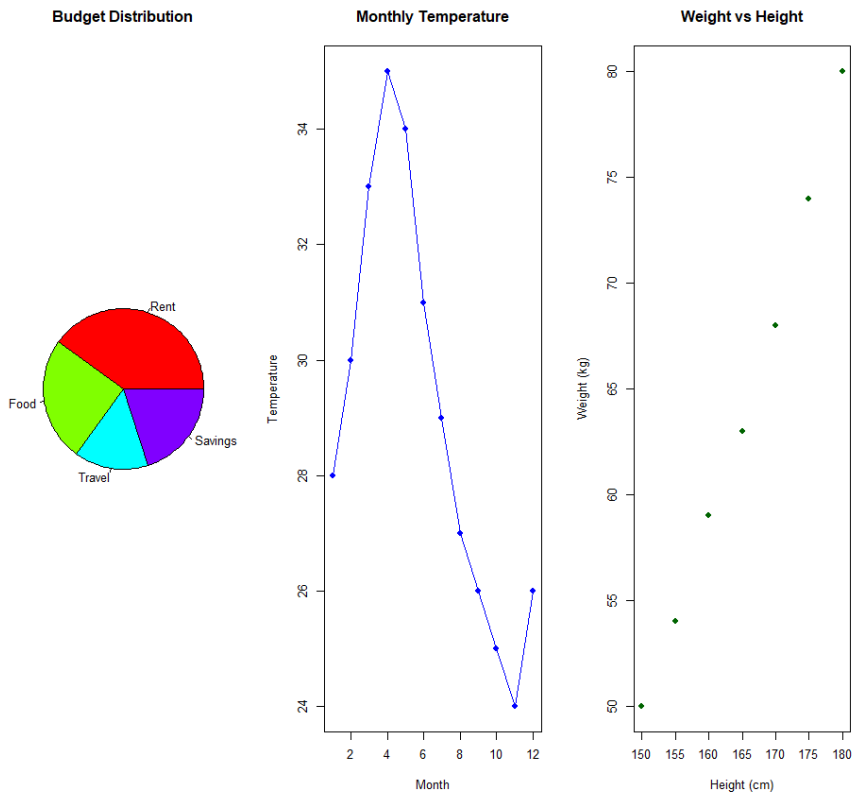
pie(budget, main = "Budget Distribution",
    col = rainbow(length(budget)))

plot(1:12, temp, type = "o", pch = 16,
     xlab = "Month", ylab = "Temperature",
     main = "Monthly Temperature")

plot(height, weight, pch = 19, col = "darkgreen",
     xlab = "Height (cm)", ylab = "Weight (kg)",
     main = "Weight vs Height")

par(mfrow = c(1, 1))
```

OUTPUT / OBSERVATION: Three plots appear side by side: a pie chart for budget composition, a line chart for monthly temperature and a scatter plot for the relationship between height and weight.



b) Chart family and question answered

Chart	Family	Question answered
Pie chart	Part-to-whole	How is the total budget divided among categories?
Line chart	Time series	How does temperature change from month to month?
Scatter plot	Relationship	How are height and weight associated?

c) Why a pie chart is unsuitable for monthly temperatures

A pie chart is not suitable for displaying monthly temperatures because:

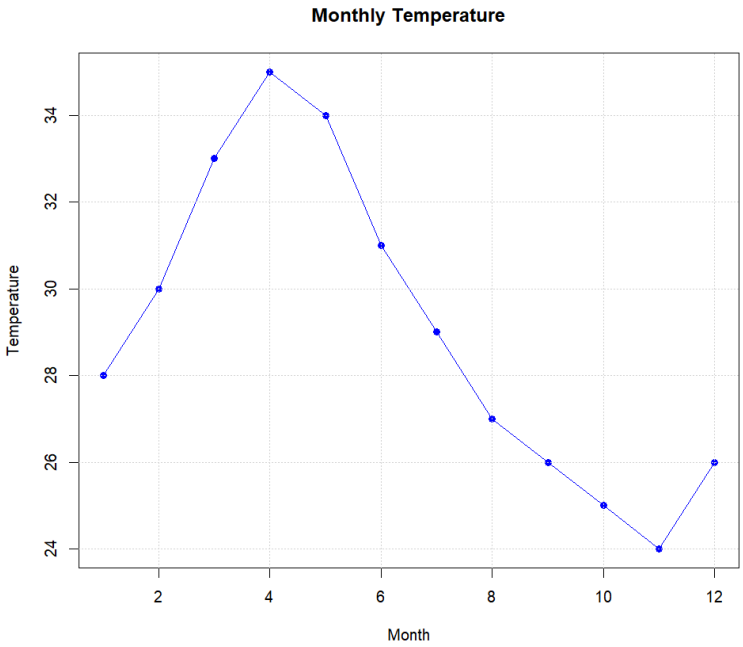
1. A pie chart represents parts of a whole.
2. Monthly temperatures are data recorded over time.
3. It does not clearly show temperature increases and decreases.
4. It makes identifying trends and seasonal patterns difficult.

Recommended Chart: Line Chart

A line chart is more suitable because:

- It displays temperature changes month by month.
- It maintains chronological order.
- It clearly shows increasing and decreasing temperatures.
- It helps identify seasonal patterns.

OUTPUT / OBSERVATION: The three chart types are displayed in a single row. A line chart is recommended for monthly temperature data



.

RESULT: The three requested charts were generated side by side and their chart families and uses were identified.

QUESTION 6: Summarising a data frame by group

AIM: To calculate department-wise mean salary, employee counts and total salary using aggregate(), table() and tapply(), and visualize the mean salaries.

a) Mean salary by department and bar chart

PROCEDURE:

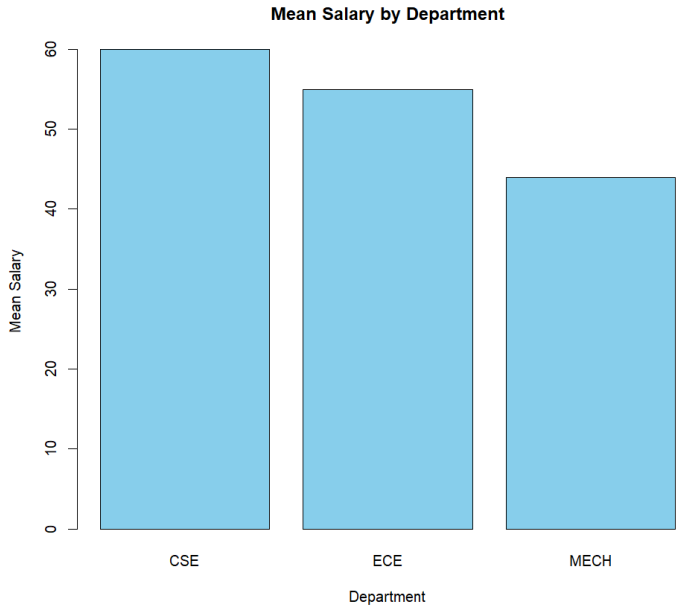
- Create the data frame with dept and salary columns.
- Use aggregate() to calculate mean salary for each department.
- Sort the resulting data frame in descending order of mean salary.
- Draw a bar chart using the sorted department names and means.

```
d <- data.frame(
  dept = c("CSE", "CSE", "ECE", "ECE", "ECE",
           "MECH", "MECH", "CSE"),
  salary = c(50, 60, 45, 55, 65, 40, 48, 70)
)

mean_salary <- aggregate(salary ~ dept, data = d, FUN = mean)
mean_salary <- mean_salary[order(-mean_salary$salary), ]
print(mean_salary)

barplot(mean_salary$salary,
        names.arg = mean_salary$dept,
        col = "steelblue",
        ylab = "Mean Salary",
        main = "Mean Salary by Department")
```

OUTPUT / OBSERVATION: Mean salary values: CSE = 60, ECE = 55, MECH = 44. The sorted bar chart ranks CSE first, ECE second and MECH third.



b) Employee count and total salary

```
# Number of employees in each department
table(d$dept)

# Total salary in each department
tapply(d$salary, d$dept, sum)
```

OUTPUT / OBSERVATION: The table() output gives employee counts by department. The tapply() output gives total salary by department.

```
> count <- table(d$dept)
>
> print(count)

 CSE  ECE MECH
   3    3    2
>
```

c) Is mean salary alone sufficient?

A bar chart of mean salary alone is not sufficient for a complete comparison. It does not show how many employees are in each department or the total salary cost. Departments with the same mean can have different employee counts. Improve the comparison by displaying mean salary together with employee count and total salary, for example in a summary table or a grouped dashboard.

```
dept <- c("CSE", "CSE", "ECE", "ECE", "ECE", "MECH", "MECH", "CSE")
salary <- c(50, 60, 45, 55, 65, 40, 48, 70)
d <- data.frame(dept, salary)
summary <- aggregate(salary ~ dept, data = d,
                     FUN = function(x) c(
                       Count = length(x),
                       Total = sum(x),
                       Mean = mean(x)
                     ))
print(summary)
```

OUTPUT / OBSERVATION: Department-wise mean salary, count and total salary are obtained. The comparison is improved by reporting all three measures.

```
>
> print(summary)
  dept salary.Count salary.Total salary.Mean
1  CSE            3          180          60
2  ECE            3          165          55
3 MECH            2           88          44
>
```

RESULT: The department data was summarized using aggregate(), table() and tapply(), and the mean salary chart was created.

QUESTION 7: Dimensions, measures and dates in Tableau

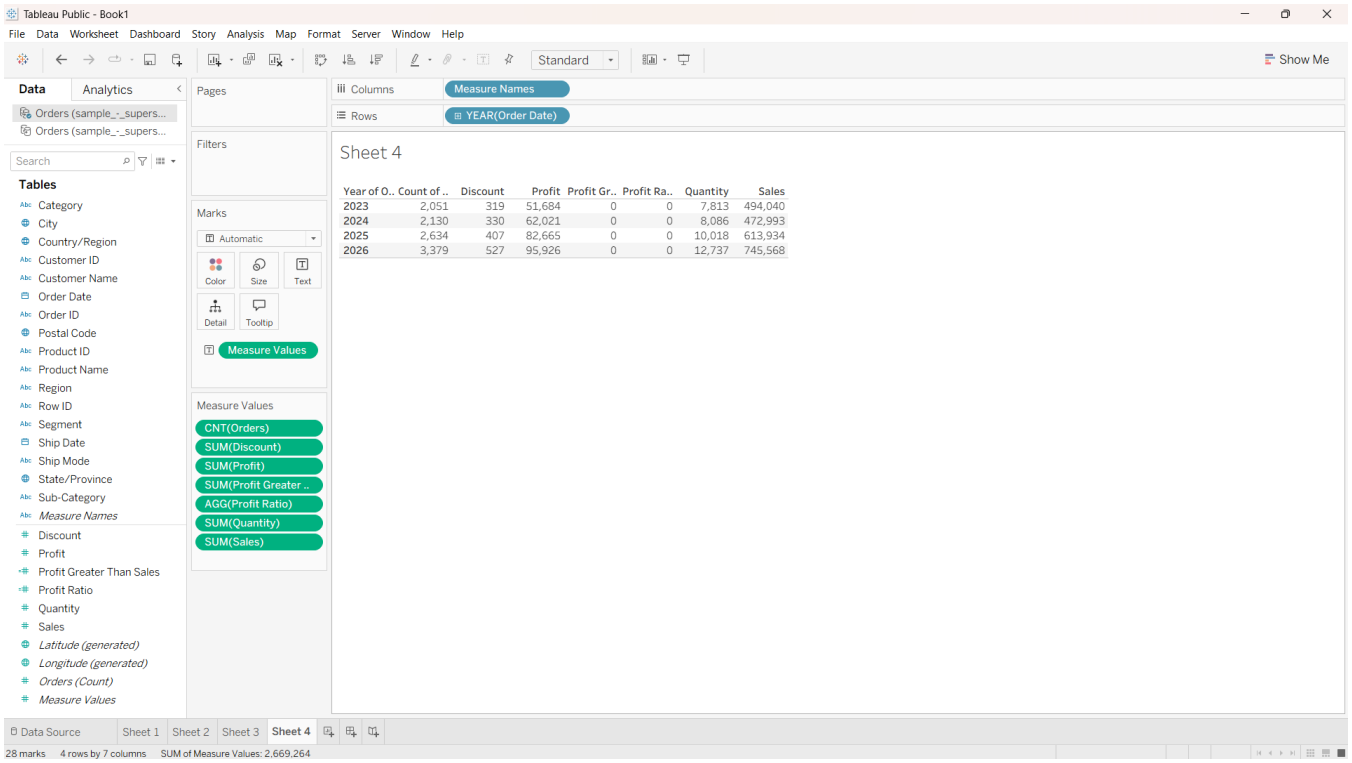
AIM: To create a year-wise Sales and Profit text table, classify dimensions and measures, and compare discrete and continuous month date representations.

a) Text table of Sales and Profit by Order Date year

PROCEDURE:

15. Connect Tableau to Retail_India_Tableau_Practice.xlsx and open Orders.
16. Open a new worksheet.
17. Drag Order Date to Rows and select YEAR(Order Date) as a discrete date part.
18. Drag Sales to Text on the Marks card.
19. Drag Profit to Text as well, or place Sales and Profit on Measure Values and Measure Names on Columns.
20. Format Sales and Profit as currency or numbers as appropriate.
21. Check that the table shows one row per year and both measures.

OUTPUT / OBSERVATION: A text table is displayed with Order Date (Year), SUM(Sales) and SUM(Profit).



b) Classify fields

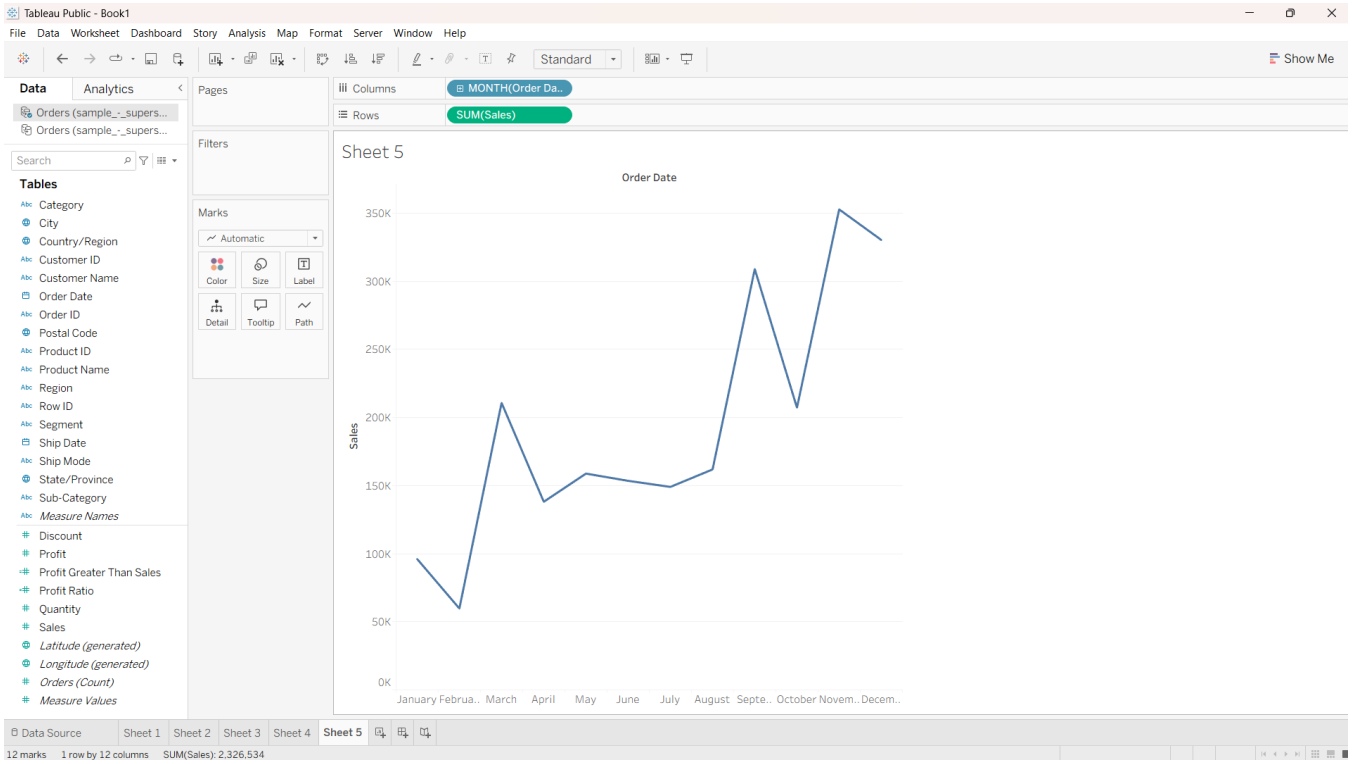
Field	Classification	Reason
Region	Dimension	Categorical field used to group records.
Segment	Dimension	Categorical customer grouping.
Order Date	Dimension (date)	Date field used to define time periods.
Sales	Measure	Numeric value that can be aggregated.
Discount offered	Measure	Numeric value that can be

		aggregated; may be used as a measure.
Row ID	Dimension	Identifier; should not be summed.

c) Discrete month or continuous month for seasonality

Discrete MONTH(Order Date) creates separate month categories (January, February, etc.) and is useful for comparing month names across years when year is also included. Continuous MONTH(Order Date) creates a continuous date axis and is useful for showing a chronological trend. For seasonality across multiple years, use month-of-year categories with Year separated or use a continuous date axis with the full month-year date, depending on the intended comparison. Avoid grouping all January observations across years if the goal is a chronological trend.

OUTPUT / OBSERVATION:



RESULT: A year-wise Sales and Profit table was constructed and field roles and date options were explained.

QUESTION 8: Sorting, filtering and Top N in Tableau

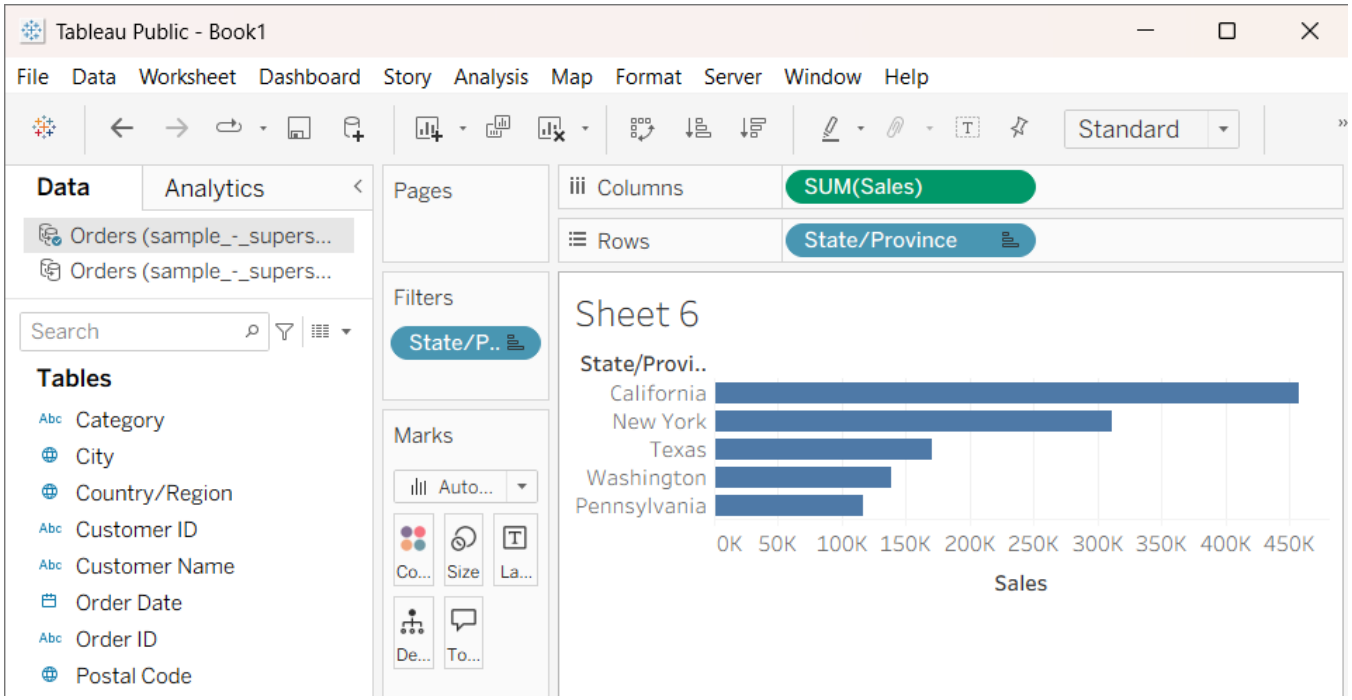
AIM: To create a descending Sales bar chart for the top five States / UTs, calculate their share of total Sales, and evaluate a fixed Top N versus a parameter.

a) Top 5 States / UTs by Sales

PROCEDURE:

- Open the Orders worksheet in Tableau.
- Drag State / UT to Rows and Sales to Columns; set the Marks type to Bar.
- Right-click State / UT and choose Filter.
- Open the Top tab, select By field, choose Top 5 by SUM(Sales), then apply.
- Sort the bars in descending order by SUM(Sales).
- Show the filter or title so the view clearly indicates Top 5.

OUTPUT / OBSERVATION: A descending bar chart displays the five States / UTs with the highest total Sales. The state names and Sales values depend on the workbook.



b) Share of total Sales

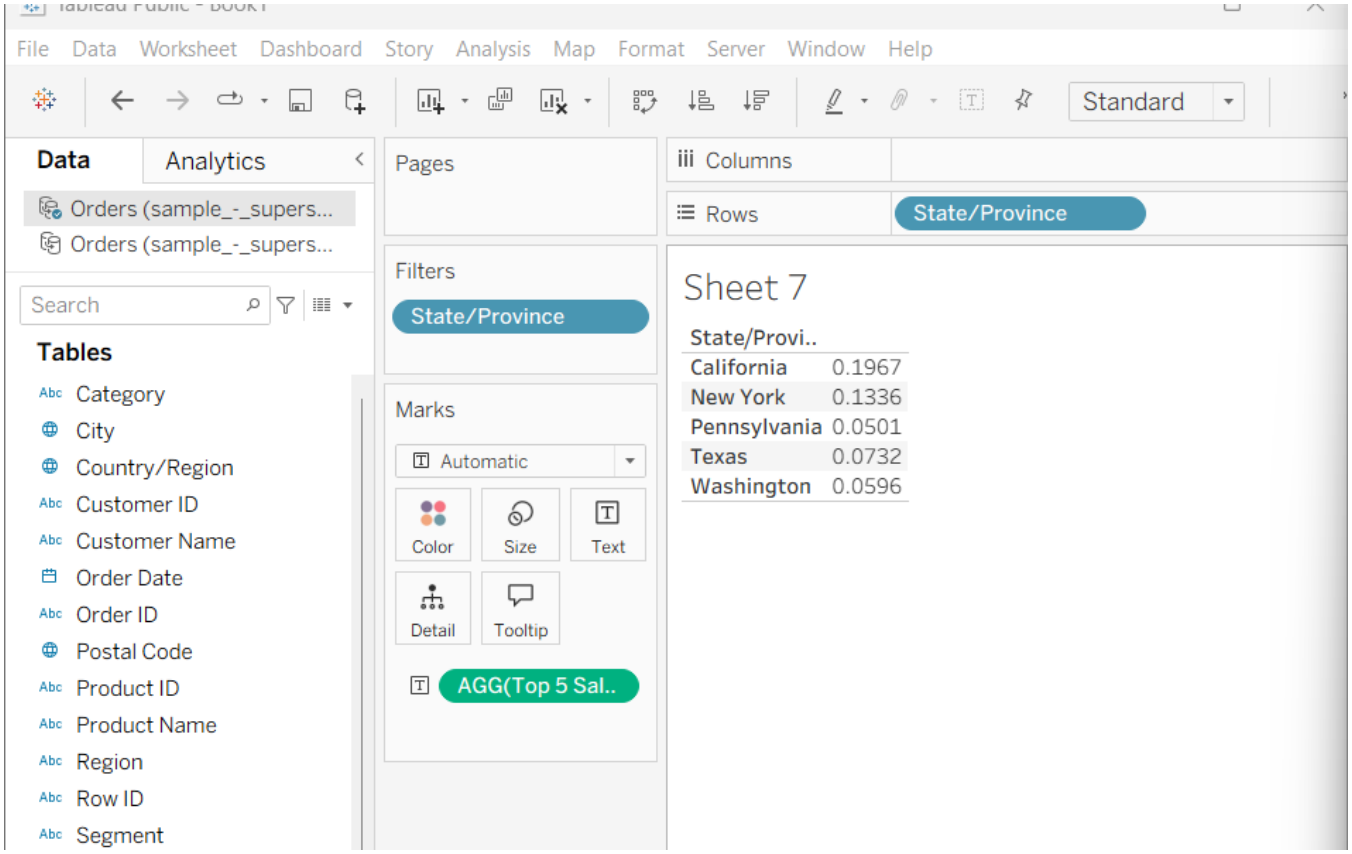
Use a table calculation or a Level of Detail expression to compare the Top 5 Sales with overall Sales. One straightforward method is to create a FIXED total for all sales, then calculate the ratio.

```
[Overall Sales] = { FIXED : SUM([Sales]) }

[Top 5 Sales Share] =
SUM([Sales]) / MIN([Overall Sales])
```

With the Top 5 filter configured as a dimension filter, verify that the overall total remains unaffected. If the filter is made a context filter, the FIXED expression may be affected; use a table calculation such as Percent of Total or carefully configure the order of operations. Format the ratio as a percentage.

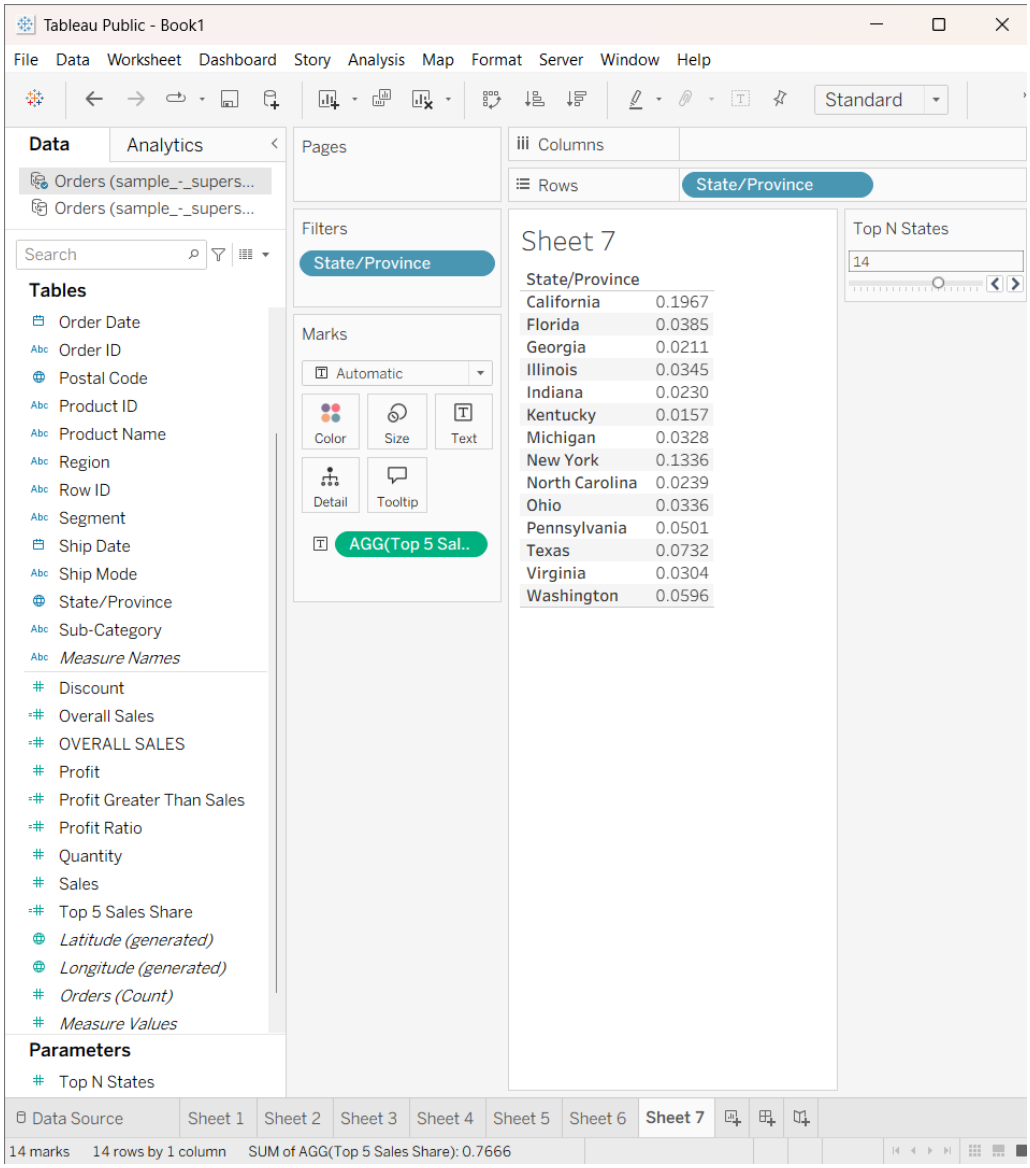
OUTPUT / OBSERVATION



c) Fixed N or parameter

A fixed Top 5 is simple and suitable when the report requirement is permanently five states. A parameter is more flexible because users can change the number of displayed states (Top 3, Top 5, Top 10, etc.) without editing the worksheet. A parameter is recommended for an interactive dashboard; use a fixed value for a one-time task with an unchanging requirement.

OUTPUT / OBSERVATION: The Top 5 bar chart and share calculation are configured. The Top N value is recommended as a parameter for reusable interactive analysis.



RESULT: A descending Top 5 state sales chart and a method to calculate its share of overall sales were prepared.

QUESTION 9: Calculated fields in Tableau

AIM: To create row-level Loss or Profit and Shipping Days calculated fields, count loss/profit lines, compare loss-making segments and distinguish row-level from aggregate logic.

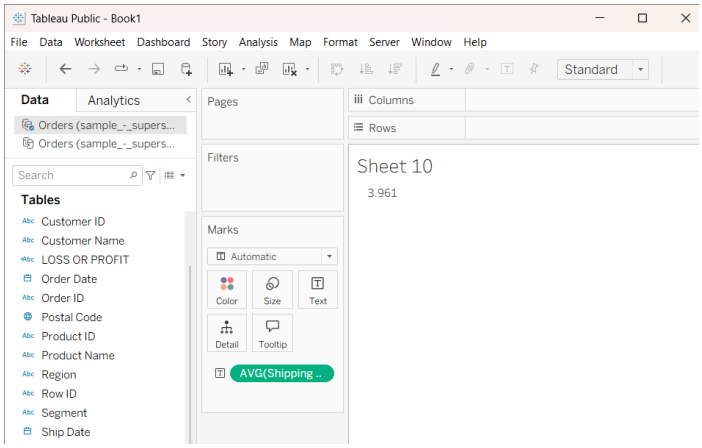
a) Create calculated fields and report line counts

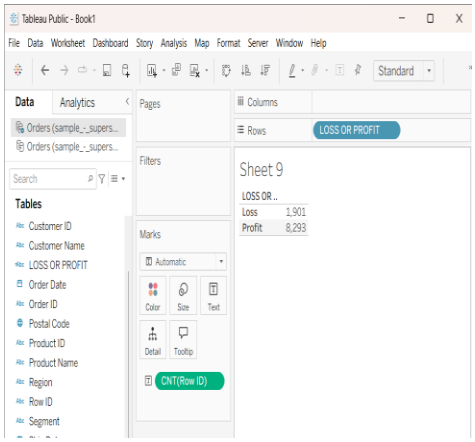
PROCEDURE:

- Open the Orders sheet in Tableau.
- Create a calculated field named Loss or Profit using the row-level formula below.
- Create a calculated field named Shipping Days using the date difference formula below.
- Place Loss or Profit on Rows and Number of Records on Text to count loss and profit lines.
- Place Shipping Days on Text and set its aggregation to Average to obtain average shipping days.
- IF [Profit] < 0 THEN "Loss"
  ELSE "Profit"
  END

  // Calculated field: Shipping Days
  DATEDIFF('day', [Order Date], [Ship Date])

OUTPUT / OBSERVATION:





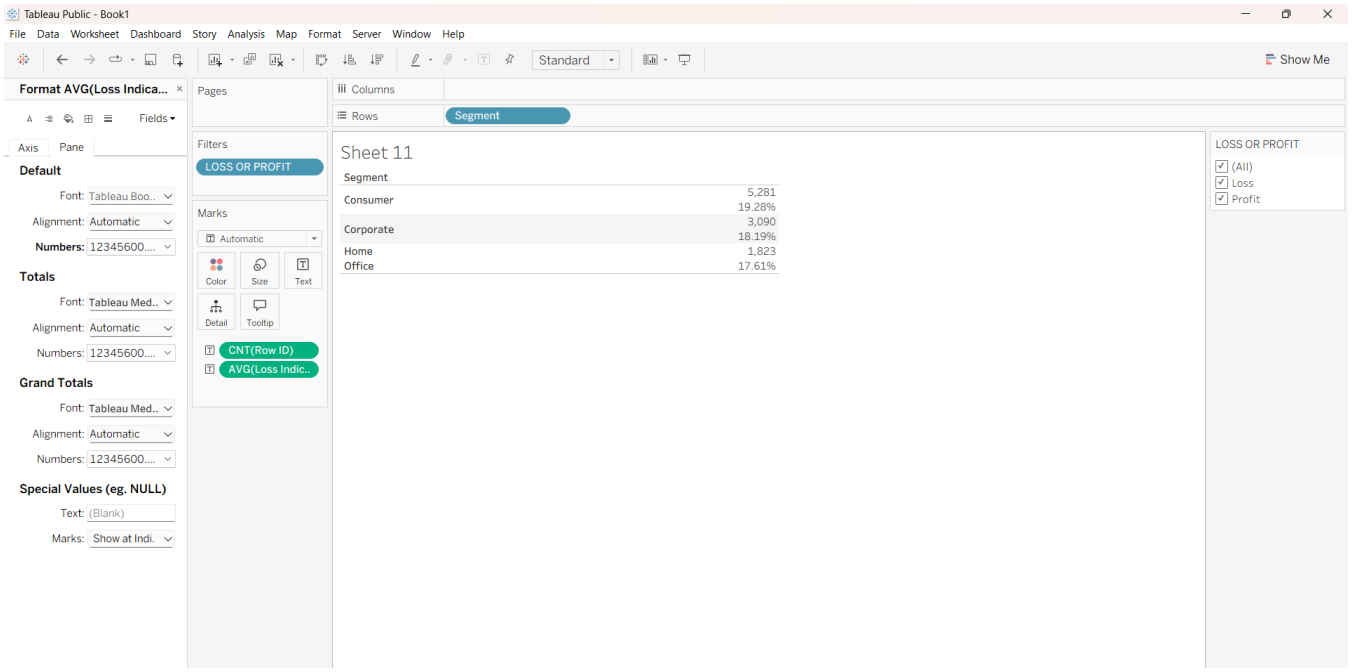
b) Share of loss-making lines in each Segment

Place Segment on Rows and Loss or Profit on Columns or Color. Use Number of Records to count lines. To calculate the loss-line share, create the following field and display it by Segment.

```
SUM(IIF([Profit] < 0, 1, 0))
/
COUNT([Row ID])
```

Format the calculated result as a percentage. If Row ID can be null or duplicated, use a suitable row-counting key or Number of Records and validate the data grain.

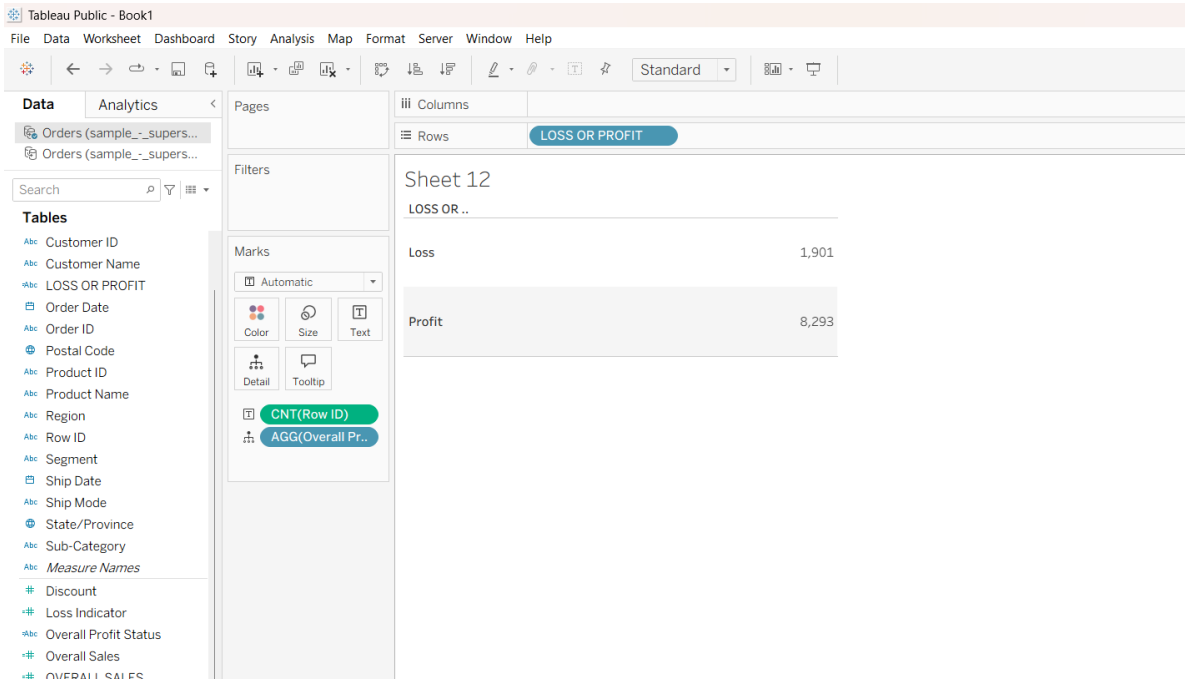
OUTPUT / OBSERVATION: A segment-wise view displays the proportion of loss-making lines.



c) Row-level IF versus aggregate IF

IF [Profit] < 0 evaluates each record separately and labels each line as loss or profit. IF SUM([Profit]) < 0 evaluates the aggregated Profit for the current view partition; it answers whether the combined profit is negative, not how many individual lines are losses. For counting loss-making lines, use row-level logic such as SUM(IIF([Profit] < 0, 1, 0)).

OUTPUT / OBSERVATION: Row-level logic is the correct choice for counting individual loss-making lines. Aggregate logic is suitable for determining whether a grouped total is a loss.



RESULT: The calculated fields and segment-level loss analysis were defined, and the difference between row-level and aggregate calculations was established.

QUESTION 10: Preparing the data in Tableau

AIM: To inspect missing values, check duplicate Row ID values, compare distinct customer counts and evaluate methods for handling missing Target values.

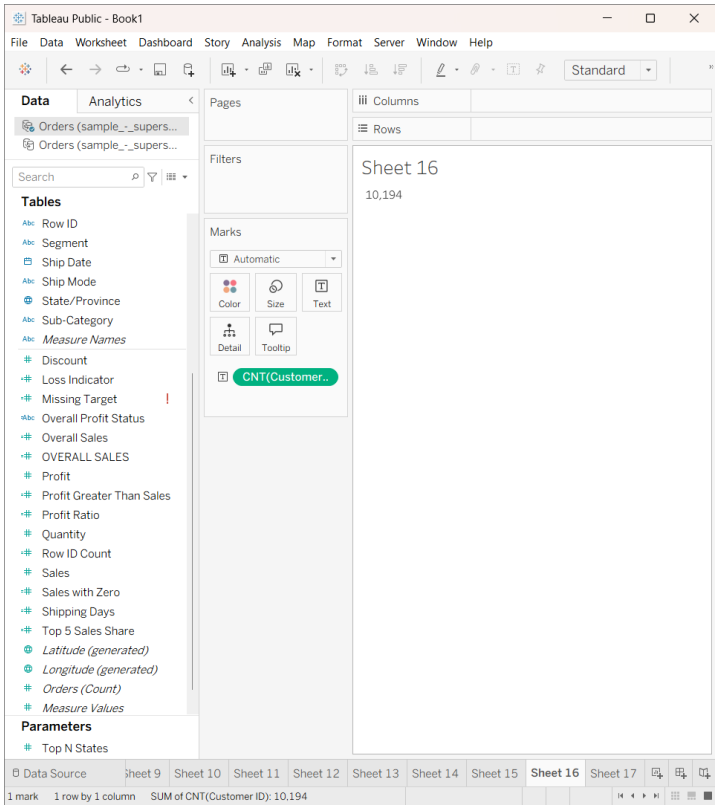
a) Count missing Target and Product Base Margin
PROCEDURE:

- Open the Orders sheet in Tableau.
- Create a calculated field named Missing Target using the formula below.
- Create a calculated field named Missing Product Base Margin using the corresponding formula.
- Place both fields on Text and aggregate using SUM to obtain the missing-value counts.
- Alternatively, use filters to isolate Null values and count the records.

```
IIF(ISNULL([Target]), 1, 0)

IIF(ISNULL([Product Base Margin]), 1, 0)
```

OUTPUT / OBSERVATION:

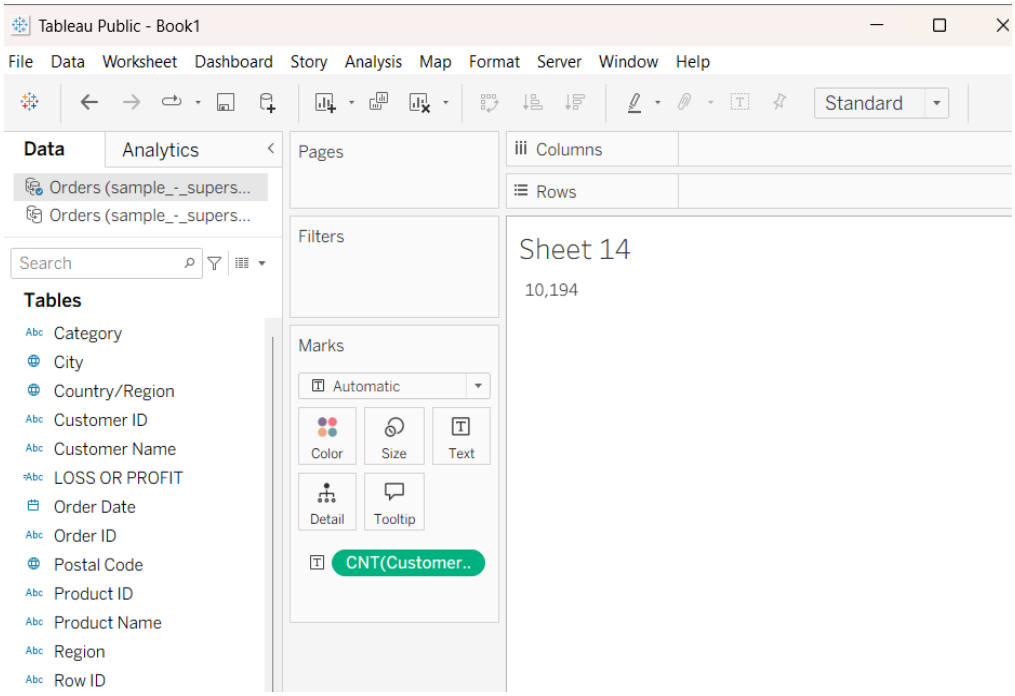


b) Duplicate Row ID and distinct customer/name counts

22. Create a worksheet and place Row ID on Rows.
23. Place Number of Records on Text or use COUNT([Row ID]) to inspect repeated IDs.
24. Filter to IDs with a count greater than one to identify duplicates.
25. Create separate distinct-count calculations for Customer No and customer Name.

```
COUNTD([Customer No])
COUNTD([Name])
```

OUTPUT / OBSERVATION

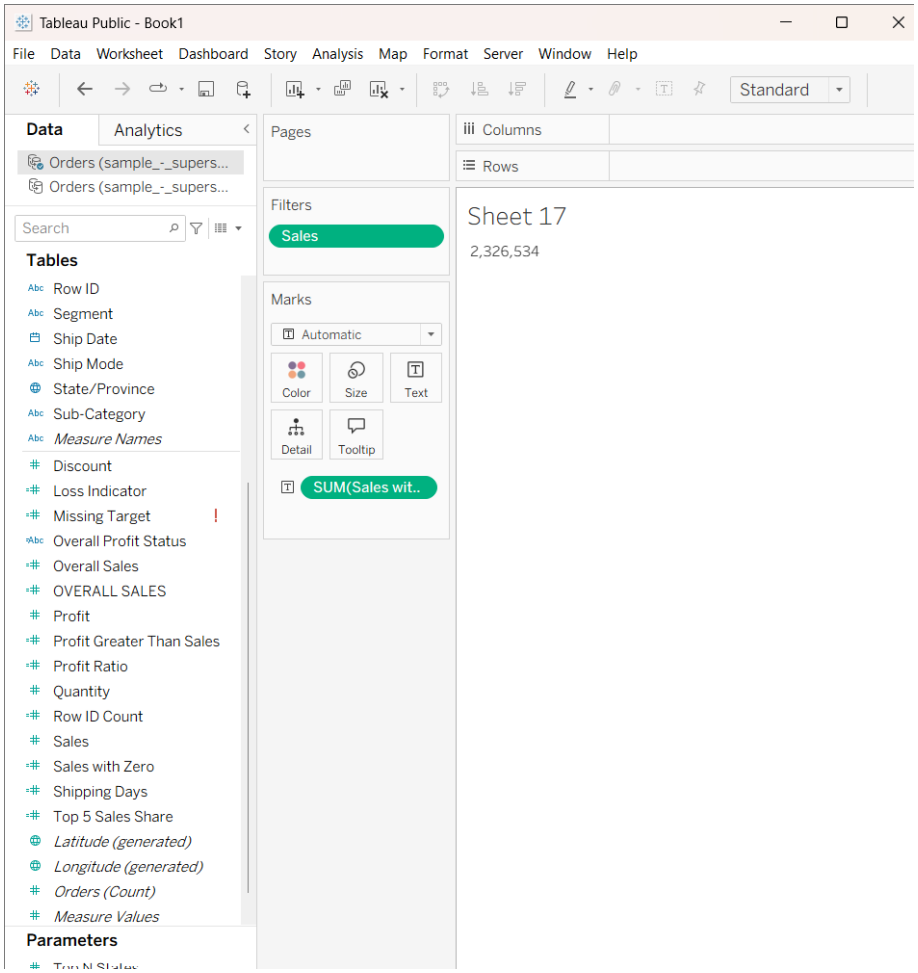


c) Choose a method for missing Target values

Method	Advantage	Risk / limitation
Filter missing rows	Keeps analysis focused on records with known Target values.	Can bias results if missingness is systematic; reduces sample size.
Replace missing with zero	Simple and retains rows.	Zero may be a real value and can distort averages, totals and models.
Delete rows	Removes incomplete records from the dataset.	Loses information and may introduce bias; should be justified.

Recommendation: Do not automatically replace missing Target values with zero unless zero is the documented meaning of a missing value. First investigate why Target is missing. For a report that requires valid Target values, filter or exclude missing records from that specific analysis and disclose the number excluded. Retain the original data and document the treatment.

OUTPUT / OBSERVATION:



RESULT: The Tableau data-preparation checks were defined, and appropriate approaches to missing Target values were evaluated.